

The Edge Grid: A Prototype for Decentralized, Verifiable Peer-to-Peer AI Inference

Harshit Raj

Dept. of CSE (IoT, Cyber-Security & Blockchain Tech.)

Sir M. Visvesvaraya Institute of Technology

Bengaluru, India

Chetan Raghuvanshi

Dept. of CSE (IoT, Cyber-Security & Blockchain Tech.)

Sir M. Visvesvaraya Institute of Technology

Bengaluru, India

Keshav Narayan

Dept. of CSE (IoT, Cyber-Security & Blockchain Tech.)

Sir M. Visvesvaraya Institute of Technology

Bengaluru, India

Mayur Agarwal

Dept. of CSE (IoT, Cyber-Security & Blockchain Tech.)

Sir M. Visvesvaraya Institute of Technology

Bengaluru, India

Guide: Dr. Savita Choudhary, Professor & Head

Abstract. Centralized inference services concentrate pricing power, availability, and trust in a single operator, and decentralized physical infrastructure networks have already shown that peer-to-peer markets can sell connectivity, storage, and general compute at scale. Selling verifiable AI inference the same way raises a problem those networks do not solve: an anonymous edge provider, operating under a hard latency budget, has both the means and the incentive to return a plausible-looking wrong answer, and naive redundancy or centralized scheduling cannot detect that cheaply. We design a network in which a job is discovered and auctioned under a sealed-bid, warm-start-aware second-price rule; executed by a streaming edge inference runtime; committed to a Merkle-backed data-availability layer; and verified by a two-tier mechanism that separates a trustless cryptographic fraud proof from a sampled, quorum-voted semantic judge, with staked collateral slashed 80/20 between reporter and treasury on confirmed fraud. We build and measure a small-scale reference implementation on commodity CPU hardware: warm time-to-first-token averaged 610 ms with a 12× cold-start penalty; a sealed-bid auction cleared correctly across 57 trials and responded near-linearly to injected per-link network delay; content-addressed model weights were fetched, independently re-verified, and correctly rejected under three tampering scenarios; and on-chain settlement conserved value exactly across all three resolution paths. The verification layer's measured behavior is the paper's central finding: the deployable, edge-hostable judge configuration caught only 30–35% of two semantically subtle fraud strategies while scoring them confidently rather than hesitantly, and a follow-up experiment shows this failure tracks judge capability rather than the judge's relationship to the model it polices, refuting an explanation this evaluation initially favored.

Keywords: decentralized physical infrastructure networks, verifiable AI inference, LLM-as-a-Judge, second-price auctions, optimistic fraud proofs, staked settlement.

I. INTRODUCTION

Centralized inference services have made large-model access cheap and reliable at the cost of concentrating pricing power, infrastructure control, and data governance in a small number of operators. Decentralized physical infrastructure networks (DePIN) already sell connectivity, storage, and general compute competitively; none solve the narrower problem this paper addresses: **selling AI inference, rather than raw compute, from anonymous edge providers under a latency budget, with correctness**

verified cheaply enough that verification does not eat the economics it protects.

Redundant execution multiplies cost by the redundancy factor for every job, not just the audited fraction, which is economically incompatible with a centralized provider's marginal cost. Centralized verification reintroduces the single point of control the network exists to remove. Reputation punishes a *pattern*, not a specific transaction. What is needed is a mechanism that prices a latency budget at routing time, verifies a *sampled* fraction of jobs cheaply, and

makes fraud's economic consequence automatic, without a central adjudicator. Two properties make this harder than verifying storage: a wrong answer need not look wrong (a fabricated fact or negated claim can be fluent and indistinguishable from a correct one), and the party best positioned to catch a subtle semantic error is itself a language model, so the verification layer inherits every reliability question that attaches to language models generally, including that it can be confidently wrong. Section 5.4 is a direct measurement of that risk.

We design the network around a strict separation between two kinds of verification claim. A **cryptographic fraud proof** establishes with certainty that a provider served output inconsistent with what it committed on chain: a fact about two hashes, needing no challenge period. A **semantic verdict** is a judgment call about whether a faithfully-delivered output is actually correct, rendered by one or more independent validator models under a quorum rule. This lets the cheap, certain tier absorb one entire class of dishonesty for a fixed cost, and confines the expensive, probabilistic tier to the dishonesty that genuinely requires judgment.

Design contributions: (1) a market protocol combining a sealed-bid second-price auction with a

warm-start handicap applied to ranking score but never to payment, since model residency measures as a near-eligibility condition rather than a tradeable quality dimension under realistic latency budgets; (2) a two-tier verification protocol with a three-valued verdict that never collapses "unreachable" into either a pass or a fraud finding; (3) settlement as an escrow whose release requires no privileged party online, with unbonding collateral that stays slashable to close a front-running window.

Prototype and evaluation contributions: (1) a working, reproducible reference implementation with every figure traceable to a configuration-snapshotted run; (2) measurement that the deployable verification configuration is confidently wrong on two of four fraud strategies, and that this tracks judge *capability* rather than *lineage*, refuting an explanation we initially favored; (3) measurement that quorum aggregation reduces false positives well below individual members at no cost to recall; (4) identification of a methodological confound: a model-generated "honest" control is not reliably honest and bounds every LLM-as-a-Judge precision figure, a problem that recurred in our own pipeline even after being diagnosed once.

II. RELATED WORK

DePIN networks already sell compute, storage, or connectivity through token-incentivized provider sets: Bittensor rewards ML work but allocates reward predominantly by stake rather than output quality [1]; Akash [2] and Golem [3] auction general compute without an inference-specific verification layer; Gensyn [4] targets verifiable ML compute oriented toward training. None couples a latency-bounded auction to a sampled semantic verification layer with staked slashing.

Morpheus [5, 6] is the closest architectural precedent: an Arbitrum-L2 marketplace where providers bid and contracts match them to serve LLM inference. We differ in mechanism (a warm-start-aware second-price auction with hard-constraint eligibility) and in adding a two-tier verification layer outside Morpheus's stated scope. DGrid [7] and Parallax [8] describe decentralized inference serving without the auction or verification layers evaluated here; both are vendor litepaper/preprint, not peer-reviewed. **PolyLink** [9] is the closest refereed

academic parallel for settlement but does not evaluate an auction's response to injected latency (§5.3).

Our data-availability layer implements Celestia's *binding* property [10] against a local store, explicitly declining its *availability* guarantee. Settlement's challenge-window structure follows the fraud-proof lineage established for rollups [11, 10, 12]. The LLM-as-a-Judge paradigm [13, 14, 15] underlies our semantic tier; our contribution is a measured finding that judge *capability*, not *lineage* relative to the policed model, determines whether subtle falsehoods are caught (§5.4), qualifying the intuition that same-family judges share blind spots. Petals [16, 17] addresses collaborative multi-peer inference, a complementary problem; Navigator [18] addresses decentralized *scheduling* alone, without verification or settlement.

No system surveyed combines Kademlia discovery, a warm-start-aware second-price auction under a hard latency budget, a two-tier verification pipeline separating a trustless fraud proof from a

capability-sensitive semantic check, and staked slashing, in one evaluated reference implementation.

III. SYSTEM DESIGN

System model. Four roles (requesters, providers, validators, and a settlement layer) compose a five-stage pipeline (discovery, market, inference, verification, settlement) communicating through a fixed message contract (Figure 1).

Threat model. Cheating providers are split into two trust classes: an output inconsistent with what was committed on chain is *provably* dishonest (a hash mismatch, checkable by anyone); an output that is faithfully delivered but factually wrong requires a judge, whose error rate is an irreducible cost. Colluding provider/validator pairs are mitigated structurally: deterministic, unpredictable sampling means a provider cannot know which jobs will be audited, but validator collusion at scale is **not** claimed solved. Sybil resistance is economic (identity multiplication is cheap, but acting under an identity without stake is not) rather than identity-based, and its adequacy against a well-resourced adversary is an open question. Latency misreporting is deferred to the general verification path rather than a dedicated mechanism, a stated gap.

Discovery and identity. One secp256k1 keypair doubles as network identity and settlement wallet. A slow-changing **capability record** (models served, tier, stake) propagates through a DHT; a fast-changing **liveness signal** (reachability, which models are *loaded and ready*) propagates directly between peers, kept separate because collapsing them forces a choice between an expensive high-frequency DHT update or a stale signal an auction that clears in seconds cannot use.

Market protocol. A sealed-bid, second-price *procurement* auction: providers bid the price they want to be paid; the lowest eligible bid wins but is paid the second-lowest eligible price, making truthful bidding

dominant. Eligibility (tier, latency, price ceiling) is checked as hard constraints before any price comparison. A warm-model discount is applied to *ranking score* only, never to payment, preserving individual rationality, because over realistic latency budgets, a cold provider's fixed load cost makes warmth closer to an eligibility condition than a tradeable quality dimension (confirmed empirically in §5.2).

Inference contract. A narrow, runtime-agnostic contract: stream tokens, report true time-to-first-token and a real per-request token count, and report cold/warm state. A provider that cannot report these honestly cannot participate.

Verification protocol. Every job is committed to a content-addressed store. A deterministic-but-unpredictable sample is audited: first a purely mechanical check (does the committed output match the store, under a Merkle proof requiring no model or judgment: a **fraud proof**, treated as certain); only if that passes does one or more validators render a semantic verdict, required to be one of three values (pass, fail, or **unavailable**), so a judge outage is never conflated with either an acquittal or a slash. Validator *diversity* is recorded as a first-class property rather than assumed.

Settlement. A fixed challenge window follows commitment; uncontested release requires no privileged party online. Confirmed fraud (proof or FAIL verdict) returns escrow to the requester and slashes provider collateral 80/20 between reporter and treasury (Figure 3). Withdrawn collateral stays slashable during an unbonding delay, closing a front-running window.

IV. IMPLEMENTATION

Five cooperating tracks share one wire contract, in Python except settlement: discovery/market over a Kademlia+GossipSub library (py-libp2p); inference against a streaming open-weight runtime exposing token-level timing (Ollama, replaceable behind the same contract); settlement as four Solidity contracts on a local development chain; weight distribution against a real IPFS daemon with client-side content-identifier recomputation (never trusting the daemon's own

claim). Every message carrying economic consequence is signed and independently re-verified.

TABLE I. DESIGN VERSUS PROTOTYPE

Component	Prototype	Production target	Limitation left open
Settlement chain	Local single-node dev chain	Arbitrum Stylus rollup	No fee market, no finality delay; fraud-proof

Component	Prototype	Production target	Limitation left open
			censorability untested
Data availability	Local Merkle-committed blob store	Celestia (DA sampling)	Binding is real; availability under deletion is not
Inference runtime	CPU-only, single 2B model	GPU-accelerated serving stack	No figure bears on GPU-tier economics
Verification judge	Small locally-hosted (deployed); larger hosted (measured separately)	Locally-hostable, adequately-capable pool	Deployable config fails adversary-favored strategies; the config that

Component	Prototype	Production target	Limitation left open
			succeeds isn't edge-hostable
Economic stake	Test-denomination collateral	Real, market-priced	No conclusion about deterrent adequacy

Every experiment ran on one host (16 cores, ~31 GB RAM, no accelerator); the auction was also run in per-container network namespaces with `tc netem`-injected link delay, isolating a per-peer link delay that a shared-loopback topology cannot express. Every run writes a timestamped, never-overwritten directory with a full config snapshot and git SHA; a component that cannot be reached raises a named exception rather than a plausible-looking default.

V. EVALUATION

A. Latency

Warm time-to-first-token: mean 609.6 ms, median 587.9 ms, p95 723.6 ms (n=20; **all 20 under one second**). Cold TTFT averaged 7,963.8 ms against a paired warm 653.7 ms, a **12.18×** ratio, attributable almost entirely to model *load* time (7,418 ms cold vs. 568 ms warm), not generation, the direct empirical justification for pricing warmth near eligibility rather than as a smooth quality signal.

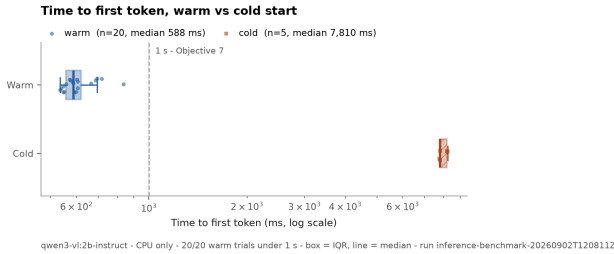


Figure 1. Time to first token, warm versus cold start, on a logarithmic time axis.

B. Auction Convergence

Fifty-seven single-machine auctions (3/4/5 nodes) cleared with zero failures; broadcast-to-award was pinned at ~2,008 ms by the fixed bid window (not a scaling signal), while un-pinned bid-arrival times carried the real signal: last-bid latency rose from 21.3 → 32.6 → 36.7 ms. Under container network namespaces with injected one-way link delay (0/10/25/50 ms), first-bid arrival rose 6.0 → 44.5 → 71.0 → 114.0 ms, close to but above a bare round-trip reference line, roughly **2.06 ms of added delay per ms injected**, consistent with GossipSub relaying through a third peer on some messages.

C. Verification Accuracy

100 trials (80 fraudulent across four corruption strategies, 20 honest) against TruthfulQA. The judge failed **none** of 20 honest answers (100% precision, 0% FPR), a corrected result: an earlier run had shown a 75% false-positive rate, since diagnosed as a fault in the honest-answer *generator*, not the judge. **Overall recall was 65%, sharply bimodal**: 100%/95% on off-topic/fabricated-entity fraud, versus **35%/30%** on plausible-substitution/negation fraud, scored *confidently* (3.80–4.05/5 against a threshold of 3), not near-threshold, ruling out "raise the threshold" as a fix.

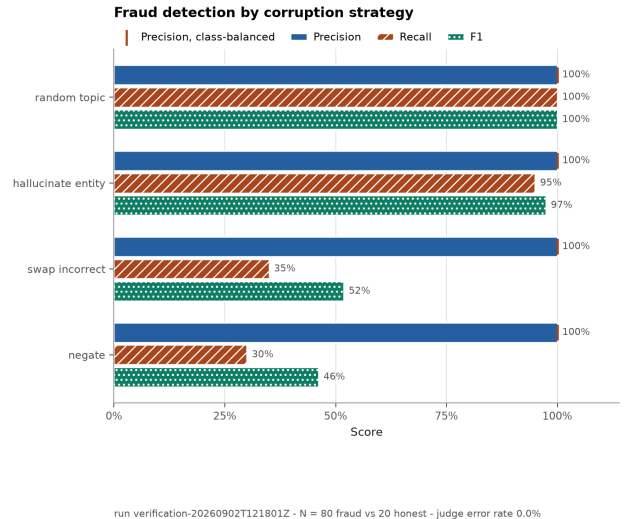


Figure 2. Fraud detection by corruption strategy: precision, recall, and F1.

Capability, not lineage. Re-judging the same 80 strings with larger hosted models: a ~13×-larger *same-family* judge recovered negation recall to 100% (98%

overall); an *unrelated* family reached 100%/90% (96% overall), refuting our own initial hypothesis that lineage, not capability, explained the gap. Under majority-vote quorum across all four panel members, recall reached 97% at a 7% false-positive rate versus 26%/16% for individual complete members, direct support for quorum aggregation, qualified by two of four members being rate-limited into near-uselessness. Judge self-consistency under paraphrase (small judge, 8 questions): a 25% verdict-flip rate on reworded claims.

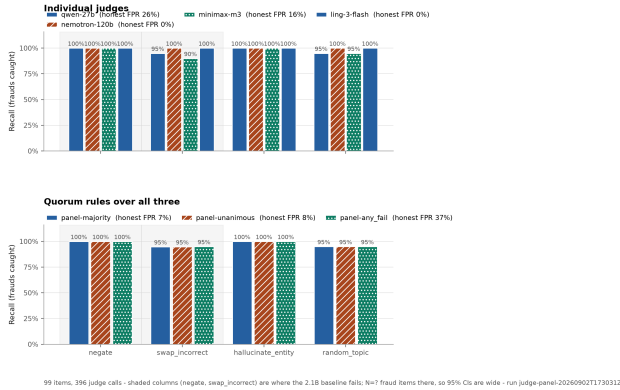


Figure 3. Recall by corruption strategy for each judge and quorum rule.

D. Weight Distribution and Cost

Five artefacts (64 KiB–48 MiB) via a real IPFS daemon: cold fetch 6.6–317.5 ms, warm (cached) under ~ 1.5 ms, speedups of **12.3×–895.8×**, with the content identifier independently recomputed on every fetch and three tampering scenarios correctly rejected against one honest control accepted. Three on-chain settlements (honest-release, fraud-proof-slash, verdict-slash) all resolved correctly with an exact 80/20 split and exact value conservation; gas ranged 32,317 (withdrawal) to 221,353 (the trustless fraud-proof, the best value in the system, since it requires no model). Composed cost: **\$0.00115** per 1,000 tokens (verification 4.8% of that) against a **\$0.002** centralized baseline, a notional-rate cost *model*, not a market observation.

VI. DISCUSSION

The prototype convincingly shows that the five-stage pipeline of Section III holds together as a running system rather than as a set of separately-plausible components: a real signed job travels from broadcast, through a real gossip-mesh auction that clears correctly and is individually rational on every measured award, through a real streaming inference call whose warm latency clears a sub-second target, through a real content-addressed weight fetch a node can verify without trusting its source, through a real Merkle-committed data-availability check, to a real on-chain escrow that resolves correctly down all three of its state-machine paths and conserves value exactly. That composition, not any individual mechanism (all of which are prior art, §II), is what this evaluation is actually able to support at small scale. What it does not show is comparably important: every network measurement, including the auction under injected latency, ran on one physical kernel; nothing here establishes behavior across real wide-area links, NAT traversal, packet loss, or peer churn; three and five node counts cannot establish a scaling law; gas figures are priced against no fee market or contention; and

staked collateral in every run is a test value with no real cost to lose, so no conclusion follows about deterrent adequacy: the mechanism's incentive-compatibility argument is a property of its construction, not something measured here.

The **central finding** is not a number but a corrected explanation: it would have been easy to conclude that a same-family judge inherits the fraud-generating model's blind spots, making diversity the fix. The panel experiment refutes this: capability, not lineage, closed the gap. A second, general finding travels alongside it: **every false-positive rate in this evaluation is confounded by the fallibility of whichever model generated the "honest" control**, a milder recurrence of the exact failure that produced the earlier 75%-false-positive result. Judge self-consistency under paraphrase compounds both: a single verdict is not a stable basis for an irreversible slash. Automated, on-chain slashing driven by an LLM's judgment therefore makes *which judge, at what capability, under what quorum rule* a safety-critical configuration choice, not a tuning parameter.

VII. CONCLUSION AND FUTURE WORK

Five mechanisms studied separately in prior literature (P2P discovery, an auction market, streaming

edge inference, two-tier verification, and staked settlement) compose into one measured, end-to-end

pipeline. The prototype's most useful result is a refuted hypothesis: the verification layer's failure on adversary-favored fraud tracks judge capability, not lineage, correcting our own initial explanation rather than confirming it.

Future work, each tied to a named limitation: (1) a genuine multi-machine deployment, the largest unaddressed threat, since every network measurement here shares one kernel; (2) a capability-adequate, *edge-hostable* judge, since the configuration that closes the recall gap is not deployable on the hardware

tier this network targets; (3) a human-adjudicated honest control set, the precondition for any clean judge-precision measurement; (4) the production substitutions of Table I (Celestia, a real rollup, GPU-accelerated serving, a public IPFS swarm); (5) an adaptive adversary that reallocates toward undetected fraud strategies; (6) zero-knowledge verification of model execution, which would make judge capability moot but remains orders of magnitude too expensive for a sub-second latency target today.

References

- [1] E. Lui and J. Sun, "Bittensor protocol: The Bitcoin in decentralized AI?" *MARBLE 2025*, Springer, pp. 145–165.
- [2] G. Osuri and A. Bozanich, "AKT: Akash network token & mining economics," Whitepaper, 2020.
- [3] Golem Factory GmbH, "The Golem project: Crowdfunding whitepaper," 2016.
- [4] Gensyn AI Ltd., "Gensyn litepaper," 2022.
- [5] Morpheus, Trinity, Neo, "Morpheus: A network for powering smart agents," Whitepaper, 2023.
- [6] MorpheusAIs, *Morpheus-Lumerin-Node* [Software], 2024-.
- [7] DGrid.AI, "DGrid AI: The decentralized AI inference network," Litepaper, 2025.
- [8] C. Tong et al., "Parallax: Efficient LLM inference service over decentralized environment," arXiv:2509.26182, 2025.
- [9] H. Liu et al., "PolyLink: A blockchain based decentralized edge AI platform for LLM inference," *IEEE Blockchain 2025*, pp. 101–108.
- [10] M. Al-Bassam et al., "Fraud and data availability proofs: Detecting invalid blocks in light clients," *FC 2021*, LNCS 12675, pp. 279–298.
- [11] L. Bousfield et al., "Arbitrum Nitro: A second-generation optimistic rollup," Offchain Labs, Whitepaper, 2022.
- [12] J. Teutsch and C. Reitwießner, "A scalable verification solution for blockchains," TrueBit Whitepaper, 2017.
- [13] L. Zheng et al., "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena," *NeurIPS 2023*.
- [14] S. Lin et al., "TruthfulQA: Measuring how models mimic human falsehoods," *ACL 2022*, pp. 3214–3252.
- [15] W.-L. Chiang et al., "Chatbot Arena: An open platform for evaluating LLMs by human preference," *ICML 2024*, PMLR 235.
- [16] A. Borzunov et al., "Petals: Collaborative inference and fine-tuning of large models," *ACL 2023 Demos*, pp. 558–568.
- [17] A. Borzunov et al., "Distributed inference and fine-tuning of large language models over the Internet," *NeurIPS 2023*.
- [18] Y. Yang et al., "Navigator: A decentralized scheduler for latency-sensitive AI workflows," *IEEE EDGE 2024*, pp. 35–47.